

Machine Design Considerations for an MHF/SPB-Converter Based Electric Drive

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Abstract—This paper identifies suitable phase/slot/pole combinations for fractional-slot concentrated winding (FSCW) permanent-magnet machines when forming an integrated electric drive together with a modular high frequency (MHF) converter or a stacked polyphase bridge (SPB) converter.

Index Terms—Electric machines, permanent magnet, fractional-slot, winding factor, integrated.

I. INTRODUCTION

By integrating the power electronic converter and electric machine into a compact, single unit, substantial advantages including reduced amount of cabling as well as reduced electromagnetic interference (EMI) can be realized [1], [2]. Obviously, an area where these advantages are especially valued is automotive traction. Relatively recently, two concepts based on the series connection of a number of low voltage, single- or three-phase converter *submodules* have been proposed for integrated electric drives in automotive applications. In the modular high-frequency (MHF) converter [3], each submodule comprises of a single H-bridge and a boost converter step as depicted in Fig. 1. In the stacked polyphase bridges (SPB) converter concept depicted in Fig. 2, each submodule comprises of a conventional three-phase bridge [4].

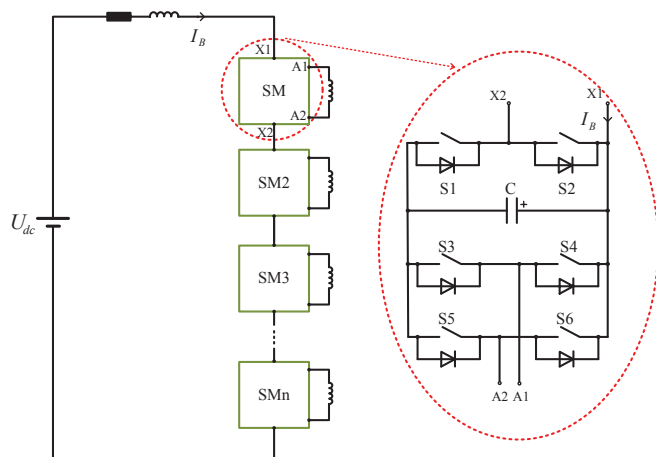


Fig. 1. The MHF converter concept and submodule configuration [3].

An advantage with both of these concepts compared to a single three-phase bridge switching the full battery voltage is

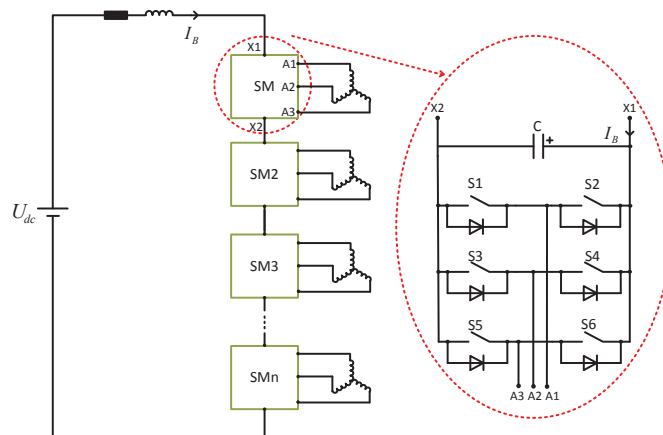


Fig. 2. The SPB converter concept and submodule configuration [4].

that a desired level of fault tolerance (post-fault output power) can be realized provided that the dc side of the faulted submodule is properly short-circuited. The number of converter submodules can be chosen freely to comply with system specifications such as battery and transistor voltage ratings and post-fault operating power. When combining the above converter concepts with a fractional-slot concentrated winding (FSCW), a very compact integrated drive can potentially be realized, mainly due to the short end turns and the high fill factors that can be realized compared to conventional distributed windings. Due to the winding distribution of FSCW, the cabling can be minimized with proper placement of the converter.

With merit of flux weakening capability, FSCW machine is employed in the traction application [5]. For the safety-critical usage, the multiphase FSCW machine with single layer is designed for the aircraft electric fuel pump which displays high fault tolerance capability [6].

During the last decade or so, the theory of FSCW windings has been developed thoroughly and it is today well established how an inappropriate selection of number of phases/slots/poles can affect machine performance significantly (see [7] and its associated references). When using a FSCW machine with an MHF/SPB converter, the restrains on the number of submodules and how each submodules should connect to the

coils of the winding should also be considered which is the focus of this paper.

A. Contribution and Outline of Paper

In this paper, the theory for FSCWs is exploited to determine appropriate winding configurations suitable for both the MHF and the SPB converter concept. Permanent-magnet type machines with interior mounted rotor magnets are considered. As a result, a number of motor designs are presented for an integrated electric drive intended for a small electric vehicle.

The paper is organized as follows. In Section II, the theory for FSCWs is used to determine phase/slot/pole combinations resulting in a high winding factor, low cogging torque and low unbalanced radial forces while at the same time avoiding excessive rotor losses. Additional criteria are then added for the winding layout to be compatible with the MHF and SPB converter concepts. As a result, a number of feasible winding layouts are identified. In Section III, key figures for a number of sample machine designs with a feasible winding layout are reported. Finally, conclusions are reported in Section IV.

II. SUITABLE PHASE/SLOT/POLE COMBINATIONS

Single-layer concentrated windings, where each slot is occupied by a single coil side, have potential benefits including a very low mutual phase coupling (beneficial for fault tolerance) and a higher phase inductance (beneficial for field-weakening operation) compared to double-layer concentrated windings where each slot occupies two coil sides [7]. However, double-layer windings are considered here due to the significantly larger space of phase/slot/pole combinations that double-layer windings allow.

A. Winding Layout and Harmonic Orders Present in the Air Gap

To determine how the coils are arranged in the stator slots, the principle of *star of slots*, adapted to fractional-slot concentrated windings in [8], is adopted in this paper. The winding periodicity t_{per} can be expressed as

$$t_{\text{per}} = \text{gcd}(Q_s, p/2) \quad (1)$$

where Q_s is the number of stator slots, p the number of poles, and $\text{gcd}(a, b)$ denotes the greatest common denominator for a and b . Introducing m_{ph} as the (odd) number of phases, the winding is practically feasible only if $Q_s/(m_{\text{ph}}t_{\text{per}})$ is an integer [8].

The harmonic order ν of the resulting MMF distribution that can be present in the air gap can be expressed as $\nu = (2k-1)t_{\text{per}}$ if Q_s/t_{per} is even (k is an integer) and as $\nu = kt_{\text{per}}$ if Q_s/t_{per} is odd [8].

B. MMF Distribution

From above, the winding can be divided into t_{per} identical groups. The angular displacement of the MMF distribution due to each adjacent phase α_{ph} can be expressed as $\alpha_{\text{ph}} =$

$2\pi/(m_{\text{ph}}t_{\text{per}})$. The current i_n in phase n , $n = 1, 2, \dots, m_{\text{ph}}$, can be expressed as

$$i_n = \sqrt{2}I \cos \left[\frac{p}{2} (\omega_m t - \alpha_{\text{ph}}(n-1) - \varphi) \right] \quad (2)$$

where ω_m is the (mechanical) angular frequency of the rotor and φ is the current phase angle.

Assuming that all coils belonging to each phase are series connected, for harmonic ν , the resulting MMF for each phase n $F_{n,\nu}$ can be expressed as

$$F_{n,\nu} = \frac{n_s Q_s k_\nu i_n}{\pi m_{\text{ph}} \nu} \cos [\nu (\theta - \alpha_{\text{ph}}(n-1))] \quad (3)$$

where k_ν is the winding factor for harmonic ν and θ is an angle spanning the air gap (expressed in mechanical degrees). For harmonic ν , the resulting MMF from all phases F_ν is now found as

$$\begin{aligned} F_\nu &= \sum_n F_{n,\nu} \\ &= \sum_n \hat{F}_\nu \cos \left[\nu \theta - \frac{p\omega_m t}{2} + \left(\frac{p}{2} - \nu \right) \alpha_{\text{ph}}(n-1) + \frac{p\varphi}{2} \right] \\ &\quad + \sum_n \hat{F}_\nu \cos \left[\nu \theta + \frac{p\omega_m t}{2} - \left(\frac{p}{2} + \nu \right) \alpha_{\text{ph}}(n-1) - \frac{p\varphi}{2} \right] \end{aligned} \quad (4)$$

where

$$\hat{F}_\nu = \frac{n_s Q_s k_\nu I}{\sqrt{2} \pi m_{\text{ph}} \nu}. \quad (5)$$

Since $\alpha_{\text{ph}} = 2\pi/(m_{\text{ph}}t_{\text{per}})$, the MMF harmonics from each individual phase are only in phase if $p/2 \pm \nu$ is a multiple of $m_{\text{ph}}t_{\text{per}}$. Introducing $\text{sgn}(\nu)$ to denote the direction of rotation of each MMF harmonic ν with respect to the rotor. From (4), $\text{sgn}(\nu)$ can be expressed as

$$\text{sgn}(\nu) = \begin{cases} +1, & \text{if } \text{rem} \left(\frac{p}{2} - \nu, m_{\text{ph}}t_{\text{per}} \right) = 0 \\ -1, & \text{if } \text{rem} \left(\frac{p}{2} + \nu, m_{\text{ph}}t_{\text{per}} \right) = 0 \end{cases} \quad (6)$$

where $\text{rem}(a, b)$ denotes the remainder after the quotient a/b . Here, $\text{sgn}(\nu) = +1$ means that the harmonic ν rotates in the same direction as the rotor and $\text{sgn}(\nu) = -1$ means that the harmonic ν rotates in the opposite direction of the rotor.

Remark: Note that (12) is applicable only to harmonics ν that can be present in the air gap (defined in Section II-A).

Presented in [9] and evident from (4), relative to the rotor, the angular speed of the harmonics ω_ν can be expressed as

$$\omega_\nu = \omega_m \left(\frac{p}{2\nu} \text{sgn}(\nu) - 1 \right). \quad (7)$$

C. Winding Factor for Harmonic ν

For harmonic ν , the winding factor k_ν can be obtained by numerically determining the harmonic content of the MMF distribution [10] or it can be obtained by summing the phasors originating from the star of slots as outlined in [11]. Below, an analytical formula for k_ν is presented that is valid for all harmonics ν that can be present in the air gap (as defined in Section II-A) provided that the quotient Q_s/p equals or is close to an integer.

First, we introduce $q_{ph} = Q_s / (m_{ph} t_{per})$ as the number of spokes per phase in the star of slot (see [8] for more details on the principle of star of slots). From the resulting winding layout, if q_{ph} is *even*, then a set of $q_{ph}/2$ coils belonging to the same phase are evenly distributed, with an alternating sign, each separated with a distance of $y_q = \text{round}(Q_s/p)$ slots. If q_{ph} is *odd*, then a set of q_{ph} coils belonging to the same phase are evenly distributed with an alternating sign, each separated with a distance corresponding to y_q slots. Thus, for harmonic ν , the *electrical* angle between the electromotive force of two adjacent coils (both belonging to the set of $q_{ph}/2$ or q_{ph} coils) $\alpha_{ph,\nu}$ can now be expressed as

$$\alpha_{ph,\nu} = \pi - \frac{2\pi y_q \nu}{Q_s}. \quad (8)$$

Similarly to conventional distributed windings, the winding distribution factor for harmonic ν $k_{d,\nu}$ can now be expressed as

$$k_{d,\nu} = \begin{cases} \frac{\sin\left(\frac{q_{ph}\alpha_{ph,\nu}}{2}\right)}{q_{ph} \sin\left(\frac{\alpha_{ph,\nu}}{2}\right)}, & \text{if } q_{ph} \text{ is odd} \\ \frac{\sin\left(\frac{q_{ph}\alpha_{ph,\nu}}{4}\right)}{\frac{q_{ph}}{2} \sin\left(\frac{\alpha_{ph,\nu}}{2}\right)}, & \text{if } q_{ph} \text{ is even.} \end{cases} \quad (9)$$

For harmonic ν , the pitch factor $k_{p,\nu}$ can be expressed as [12]:

$$k_{p,\nu} = \sin\left(\frac{\pi \nu}{Q_s}\right). \quad (10)$$

Finally, the winding factor for harmonic ν can be expressed as

$$k_\nu = k_{d,\nu} k_{p,\nu}. \quad (11)$$

The expression for $k_{d,\nu}$ given in (9) is valid only for winding layouts where the group of coils belonging to the set of $q_{ph}/2$ or q_{ph} coils are perfectly evenly distributed, each coil separated by y_q number of slots. This is the case provided that the equality below is true (for which $y_q = \text{round}(Q_s/p)$):

$$\text{round}(Q_s/p) = \begin{cases} \frac{\text{round}\left(\frac{Q_s q_{ph}}{p}\right)}{q_{ph}}, & \text{if } q_{ph} \text{ is odd} \\ \frac{\text{round}\left(\frac{Q_s q_{ph}}{2p}\right)}{\frac{q_{ph}}{2}}, & \text{if } q_{ph} \text{ is even.} \end{cases} \quad (12)$$

Two sample winding layouts with $y_q = 1$ and $y_q = 2$ are depicted in Fig. 3.

D. Rotor Losses

For each harmonic ν we assume that the induced rotor eddy-current losses $P_{e,\nu}$ are proportional to the square of the magnitude of the resulting air-gap flux density harmonic $\hat{B}_\nu$. Hence, $P_{e,\nu}$ can be expressed as $P_{e,\nu} \propto \hat{B}_\nu^2 \omega_\nu^2$ where $\propto$ denotes

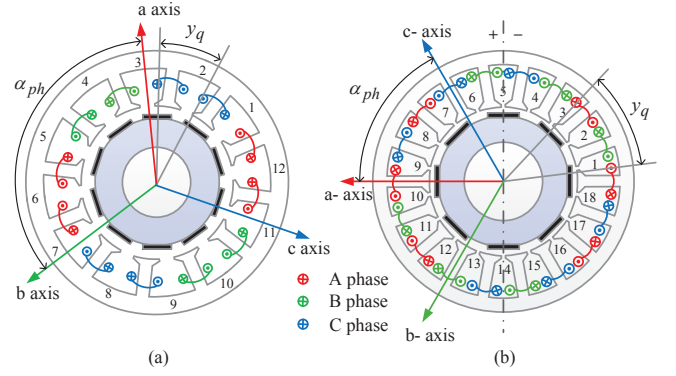


Fig. 3. Sample winding layouts for which (12) is valid: (a) $m_{ph} = 3$, $Q_s = 12$, and $p = 10$ ($y_q = 1$); (b) $m_{ph} = 3$, $Q_s = 18$, and $p = 8$ ($y_q = 2$).

‘proportional to’. In terms of the *main harmonic* $\nu = p/2$, $\hat{B}_\nu$ can be expressed as

$$\hat{B}_\nu = \frac{p}{2\nu} \frac{k_\nu}{k_{\nu=p/2}} \hat{B}_{\nu=p/2}. \quad (13)$$

From (7), the total rotor eddy-current losses P_e can be expressed as

$$P_e = \sum_\nu P_{e,\nu} \propto \underbrace{\sum_\nu \left(\frac{p}{2\nu}\right)^2 \left(\frac{k_\nu}{k_{\nu=p/2}}\right)^2 \left(\frac{p}{2\nu} \text{sgn}(\nu) - 1\right)^2}_{p_{\text{loss}}}. \quad (14)$$

The rotor loss indicator p_{loss} in (14) is introduced in [13] as an indicator for identifying combinations of Q_s , m_{ph} , and p that should not lead to excessive rotor losses.

E. Cogging Torque and Unbalanced Radial Forces

A high value of $\text{lcm}(Q_s, p)$, where lcm denotes the least common multiple, leads to a low cogging torque [14]. Hence, $\text{lcm}(Q_s, p)$ represents a useful indicator for comparing different combinations of m_{ph} , Q_s , and p .

In [15], the number $X' = \text{gcd}(Q_s, p)$ is introduced as the number of symmetries in the winding layout (X' is equal to the *vibration mode* in [10], [12]). If $X' > 1$, the radial forces in the air gap are symmetrically distributed and unbalanced radial forces do (ideally) not appear. Note that X' can also be expressed as

$$X' = \begin{cases} t_{per}, & \text{if } q_{ph} \text{ is odd} \\ 2t_{per}, & \text{if } q_{ph} \text{ is even.} \end{cases} \quad (15)$$

Table I–III include the winding factor for the main harmonic $k_{\nu=p/2}$, p_{loss} , and $\text{lcm}(Q_s, p)$ for $m_{ph} = 3$, $m_{ph} = 5$, and $m_{ph} = 7$, respectively. Combinations for which the number of slots per pole per phase q is larger or equal to one have been grayed out due to the low resulting winding factor for the main harmonic.

TABLE I
 $k_{\nu=p/2}, p_{\text{LOSS}}, \text{LCM}(Q_s, p)$ FOR Q_s/p COMBINATIONS WHEN $m_{\text{PH}} = 3$

$Q_s \setminus p$	4	6	8	10	12	14
6	0.866, 0.75, 12	N.F.	0.866, 37.0, 24	0.5, 902.3, 30	N.F.	0.5, 1777.5, 42
9	N.F.	0.866, 0.743, 18	N.F.	N.F.	0.866, 37.0, 36	N.F.
12	$q=1$	N.F.	0.866, 0.737, 24	0.933, 6.4, 60	N.F.	0.933, 21.0, 84
15		N.F.	N.F.	0.866, 0.732, 30	N.F.	N.F.
18		$q=1$	0.617, 0.789, 72	0.735, 2.1, 90	0.866, 0.723, 36	0.902, 4.655, 126
21			N.F.	N.F.	N.F.	0.866, 0.719, 42
24			$q=1$	0.583, 2.1, 120	N.F.	0.76, 2.0., 168
27				N.F.	0.617, 0.778, 108	N.F.
30				$q=1$	N.F.	0.64, 1.1, 210
N.F.	Non-feasible winding layout ($Q_s/(m_{\text{ph}}t_{\text{per}})$ is not an integer) or unbalanced radial forces ($X'=1$)					
	Suitable for MHF/SPB converter (Constraints in Table IV are fulfilled)					
	$q \geq 1$					

TABLE II
 $k_{\nu=p/2}, p_{\text{LOSS}}, \text{LCM}(Q_s, p)$ FOR Q_s/p COMBINATIONS WHEN $m_{\text{PH}} = 5$

$Q_s \setminus p$	4	6	8	10	12
10	0.588, 0.163, 20	0.809, 0.52, 30	0.951, 1.5, 40	N.F.	0.951, 14.8, 60
15	N.F.	0.588, 0.159, 30	N.F.	N.F.	0.951, 1.5, 60
20	$q=1$	0.448, 0.119, 60	0.588, 0.155, 40	N.F.	0.809, 0.502, 60
25		N.F.	N.F.	0.588, 0.151, 50	N.F.
30		$q=1$	0.410, 0.113, 120	N.F.	0.588, 0.148, 60
N.F.	Non-feasible winding layout ($Q_s/(m_{\text{ph}}t_{\text{per}})$ is not an integer) or unbalanced radial forces ($X'=1$)				
	Suitable for MHF/SPB converter (Constraints in Table IV are fulfilled)				
	$q \geq 1$				

TABLE III
 $k_{\nu=p/2}, p_{\text{LOSS}}, \text{LCM}(Q_s, p)$ FOR Q_s/p COMBINATIONS WHEN $m_{\text{PH}} = 7$

$Q_s \setminus p$	4	6	8	10	12
14	0.434, 0.071, 28	0.623, 0.192, 42	0.782, 0.437, 56	0.901, 0.944, 70	0.975, 2.014, 84
21	N.F.	0.434, 0.067, 42	N.F.	N.F.	0.782, 0.428, 84
28	$q=1$	0.328, 0.051, 84	0.434, 0.065, 56	0.529, 0.138, 140	0.623, 0.178, 84
35		N.F.	N.F.	0.434, 0.063, 70	
N.F.	Non-feasible winding layout ($Q_s/(m_{\text{ph}}t_{\text{per}})$ is not an integer) or unbalanced radial forces ($X'=1$)				
	Suitable for MHF/SPB converter (Constraints in Table IV are fulfilled)				
	$q \geq 1$				

F. Converter Integration

To enable a tight integration, the submodules of the converter should be placed closely into the corresponding stator coils. Thereby, power cables can be kept short which also is beneficial from an EMI point of view. Sample realizations of such an integration are depicted in Fig. 4.

For both the MHF and the SPB converter, for each submodule to be identical, the identity in (12) has to hold.

For the MHF converter, each submodule is connected to coils belonging to the same phase. If $q_{\text{ph}} = 1$, then combinations with $t_{\text{per}} = 1$ minimizes the number of submodules m_{sm} . If $q_{\text{ph}} > 1$, then the condition $y_q = 1$ minimizes the distance between each coil in the same phase and $t_{\text{per}} = 1$ is recommended to minimize m_{sm} . For $t_{\text{per}} = 1$, m_{sm} can then be selected as $m_{\text{sm}} = m_{\text{ph}}$ or $m_{\text{sm}} = 2m_{\text{ph}}$.

For the SPB converter, each submodule is connected to a set of three phase windings [4]. Hence, $m_{\text{ph}} = 3$ must hold. As mentioned above, if the identity in (12) holds, t_{per} groups of identical three phase windings are evenly distributed along the air gap. For combinations with $q_{\text{ph}} = 1$, each submodule will

be connected to three coils, wound in three adjacent slots, each belonging to a separate phase. The total number of submodules for the SPB converter is then $m_{\text{sm}} = t_{\text{per}}$.

The above recommended constraints for selecting suitable combinations of m_{ph} , Q_s , and p for the MHF and SPB converters are summarized in Table IV.

TABLE IV
 RECOMMENDED CONSTRAINTS FOR MACHINE-CONVERTER
 INTEGRATION

	MHF	SPB
Constr.	$\{t_{\text{per}} = 1, y_q = 1, q_{\text{ph}} > 1 t_{\text{per}} = 1, q_{\text{ph}} = 1\}$ Eq. (12) holds	$q_{\text{ph}} = 1$ $m_{\text{ph}} = 3$ Eq. (12) holds
m_{sm}	m_{ph} or $2m_{\text{ph}}$	t_{per}

G. Recommended Combinations of m_{ph} , Q_s , and p

From Table I–III and Table IV, a number of recommended combinations of m_{ph} , Q_s , and p for the MHF and the SPB converter topology are reported in Table V. The recommended combinations are selected from the set of green elements in

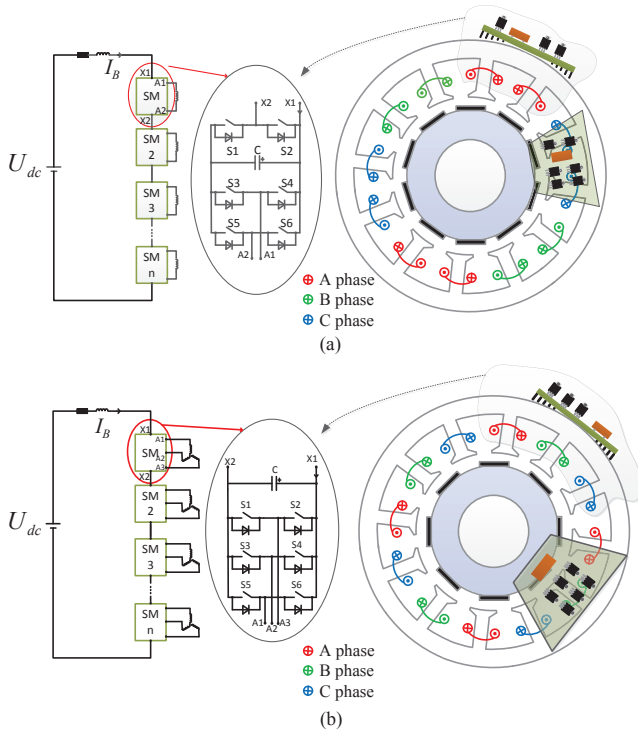


Fig. 4. Integrated drive with (a) MHF converter; (b) SPB converter.

Table I–III by a trade off between the winding factor of the main harmonic $k_{\nu=p/2}$ and the associated rotor losses p_{loss} .

TABLE V
RECOMMENDED COMBINATIONS OF m_{ph} , Q_s , AND p

m_{ph}	3		5	7
Q_s	12	18	10	14
p	8	12	6	10
Converter	SPB	SPB	MHF	MHF
m_{sm}	4	6	5/10	7/14

III. SAMPLE MACHINE DESIGNS

From the four recommended combinations of m_{ph} , Q_s and p reported in Table V, four corresponding 30 kW machine designs that fulfill the specifications for a small electric vehicle are reported in Table VI. All machines have been designed using magnets mounted interior in the rotor, i.e., a so called IPM rotor. A schematic layout illustrating how the submodules connect to each of the four machines is depicted in Fig. 5.

Remark: Phase voltage and phase current displayed are the fundamental RMS value.

IV. CONCLUSION

In this paper, a number of recommendations for suitable phase/slot/pole combinations were derived for an electric machine of FWCW type when forming an integrated electric drive with an MHF or SPB type converter. From the derived recommendations, a number of sample machine designs were

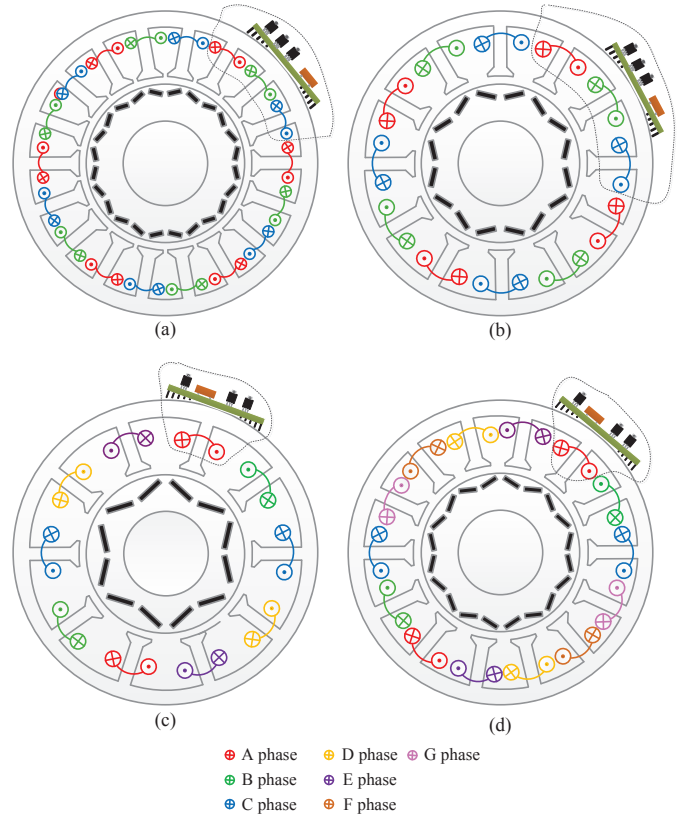


Fig. 5. Submodule connection arrangement.

TABLE VI
SUMMARY OF MACHINE PARAMETERS

Machine #		1	2	3	4
m_{ph}	-	3	3	5	7
Q_s	-	18	12	10	14
p	-	12	8	6	10
Rated power	kW	30.6	31.0	31.0	31.4
Torque	Nm	146	148	148	150
Rated speed	rpm	2000	2000	2000	2000
Phase voltage	V	163	160	71	38
Stator OD	mm	217.2	221.4	221.0	216.3
Stator ID	mm	150	157	151	154
Shaft diameter	mm	70	70	70	70
Airgap length	mm	0.5	0.5	0.5	0.5
Stack length	mm	166.7	151.7	164.5	158.1
Magnet thickness	mm	7.16	7.16	7.16	7.16
Magnet length	mm	12.0	17.4	22.9	14.2
Current rms density	A/mm ²	10	10	10	10
Slot area	mm	271.0	399.6	489.8	333.6
Phase current	A	75.3	76.9	102.0	139.0
Phase resistance	Ω	0.074	0.072	0.029	0.009
# turns per slot	-	18	26	24	12

presented that all fulfill specifications for a small electric vehicle.

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